

Statistics

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1 Mean, Median, Mode, Range

1.1 Mean

The mean is the same thing as the average. We are adding everything together, and then dividing by how many things we have.

Definition 1.1 – Mean

$$\text{Mean} = \frac{a_1 + a_2 + \cdots + a_n}{n}$$

1.2 Median

If we order all the numbers we have, the median is the middle number.

Example 1.1 What is the median of $\{47, 23, 89, 12, 56, 71, 34\}$?

The ordered list is $\{12, 23, 34, 47, 56, 71, 89\}$, so the median is 47.

If the list have an even amount of numbers, we will average the middle two numbers.

1.3 Mode

In a list of numbers, the mode is the number that appeared the most amount of times.

1.4 Range

The range is the largest number take away the smallest number in the list. Using the previous example:

Example 1.2 What is the range of $\{47, 23, 89, 12, 56, 71, 34\}$?

The range is $89 - 12 = 77$.

Problem 1.1 A lift has a weight limit of 1500kg. If the average weight of the people in the lift is 80kg and the combined weight of all people is 100kg over the weight limit, how many people are in the lift?

2 Probability

Probability will always be between 0 and 1, 1 is absolutely certain will happen, and 0 is absolutely will not happen.

Example 2.1 The probability of rolling any number between 1 and 6 when rolling a dice is 1.

To calculate a probability, we will do

$$\text{Probability} = \frac{\text{Number of desired outcomes}}{\text{Total number of possible outcomes}}$$

Problem 2.1 What is the probability that, when I roll two dice and multiply their numbers together, the answer I get is larger than 20? Express your answer as a simplified fraction.

Problem 2.2 A bag contains 5 red marbles, 3 blue marbles, and 2 green marbles. If you draw one marble without looking, what is the probability that it is blue?

Problem 2.3 A coin is flipped twice. What is the probability of getting at least one heads? Express your answer as a simplified fraction.

Problem 2.4 A deck of cards has 52 cards. If you draw one card, what is the probability that it is a heart or a diamond? Express your answer as a simplified fraction.

3 Graphs

3.1 Bar Graph

A bar graph displays categorical data with rectangular bars.

3.2 Stem and Leaf Plot

A stem and leaf plot organizes data by showing stems (tens digits) and leaves (ones digits).

3.3 Box and Whisker Plot

A box and whisker plot shows the distribution of data using quartiles. There are upper and lower quartiles.

3.4 Line Graph

A line graph shows how data changes over time by connecting data points with lines.

4 Problems

For problems 1, 2, 3, 4, 5, find the mean, median, mode and range for each of these datasets.

Problem 4.1 The heights (in cm) of 9 students are:

162, 158, 170, 165, 158, 175, 168, 158, 172

Problem 4.2 A survey recorded the number of hours 8 students studied per week:

5, 12, 8, 10, 12, 7, 12, 9

Problem 4.3 Test scores for a class are: 85, 92, 78, 88, 92, 95, 78, 88, 92, 81.

Problem 4.4 Daily temperatures ($^{\circ}\text{C}$) for one week: 18, 22, 20, 25, 22, 19, 23.

Problem 4.5 Number of books read by 10 students in a month: 3, 5, 2, 5, 4, 5, 3, 6, 5, 2.

Problem 4.6 Jake has 2 children, a boy and a girl. Each of his children has two children, a boy and a girl. If he has one child from each family to stay, what is the probability that they are both boys or both girls?

Problem 4.7 The average score of the 11 players in a cricket team is 57. One player's score was wrongly recorded as 66, when it should have been 99. What should the average score be?

Problem 4.8 In his pocket Fred has one of each of the current coins and notes in New Zealand. He takes two out of his pocket. How many different totals could he have in his hand?

Problem 4.9 An unusual die has six faces labelled 1, 2, 3, 3, 5, 7. If two of these dice are rolled, and the numbers showing on the upper faces are added, what is the number of possible different sums?